1 Open your Windows or Linux system, use the command line to assign a static IPv6 address to your primary network adapter, and verify the configuration using ipconfig (Windows) or ifconfig/ip (Linux).

Practical: Assign a Static IPv6 Address

You can perform this practical on your Windows laptop using Command Prompt (CMD).

Step 1: Check your network adapter

Open CMD as Administrator and run:

```
netsh interface ipv6 show interfaces
```

Find your primary network adapter, such as `Wi-Fi` or `Ethernet`.

Step 2: Assign a static IPv6 address

For this example, we will use the documentation-only IPv6 address `2001:db8:1::10/64`. Use this only as a lab example, not as a working public internet address.

```
netsh interface ipv6 add address "Wi-Fi" 2001:db8:1::10
```

Replace `"Wi-Fi"` with your actual adapter name. Run this command in Administrator CMD.

Step 3: Verify the configuration

Run:

```
ipconfig
```

Look under your Wi-Fi adapter for the IPv6 Address field.

You can also run:

```
netsh interface ipv6 show addresses
```

Expected result

Item	Example
Network Adapter	Wi-Fi
IPv6 Address	`2001:db8:1::10`
Prefix Length	`/64`
Verification	`ipconfig`

Important: This example address is reserved for documentation and will not provide normal IPv6 internet connectivity. For a real static IPv6 configuration, use an address and prefix assigned by your network administrator or ISP.

2 Configure a unique local IPv6 address (starting with 'fd') on your system and confirm connectivity by pinging another device on your LAN that also has a unique local IPv6 address.

Hint: Use the ping6 or ping -6 command for IPv6 testing.

Practical: Configure Unique Local IPv6 and Test Connectivity

A Unique Local IPv6 Address (ULA) starts with `fd` and is used for communication within a local network.

Step 1: Configure an IPv6 address

On Windows, open Command Prompt as Administrator and check your adapter name:

```
netsh interface ipv6 show interfaces
```

For example, assign a ULA to your Wi-Fi adapter:

```
netsh interface ipv6 add address "Wi-Fi" fd12:3456:789a::10
```

Step 2: Configure another LAN device

On the second device, assign a different address in the same `/64` subnet:

```
fd12:3456:789a::20/64
```

Both devices must be on the same LAN, and their IPv6 addresses must not conflict with existing addresses.

Step 3: Verify the configuration

On Windows, run: **p 4: Test connectivity**

```
ipconfig
```

Check that the IPv6 address appears under your network adapter.

Ste

From the first device, ping the second device:

```
ping -6 fd12:3456:789a::20
```

On Linux, you can use:

```
ping -6 fd12:3456:789a::20
```

Expected Result

Item	Example
Device 1 IPv6	`fd12:3456:789a::10/64`
Device 2 IPv6	`fd12:3456:789a::20/64`
Address Type	Unique Local IPv6
Test Command	`ping -6 fd12:3456:789a::20`
Successful Result	Reply received from the second device

Conclusion: If you receive ping replies, IPv6 communication between the two devices is working. If it fails, check the adapter configuration, subnet prefix, firewall rules, and LAN connectivity.

3 Set up a temporary IPv6 address on your system and check if it changes after a reboot or network disconnect/reconnect.

Constraint: Document the steps and the observed address behavior in a text file.

Practical: Temporary IPv6 Address Test

Steps:

1. Open Command Prompt and type `ipconfig`.
2. Note the IPv6 address of the active network adapter.
3. Disconnect and reconnect the Wi-Fi.
4. Run `ipconfig` again and check the IPv6 address.
5. Restart the computer and check the address again.

Observation :

Condition	IPv6 Address Behavior
Before reconnecting	IPv6 address was recorded.
After reconnecting	The address was checked again.
After reboot	The address was checked again.

Result: A temporary IPv6 address may change after reconnecting to the network or restarting the computer. However, it may also remain the same depending on the network configuration.

Conclusion: Temporary IPv6 addresses help protect user privacy by reducing the long-term use of the same address.

4 Simulate a Zomato-style search: Write a small script (any language) that stores an array of restaurant names and their IPv6 addresses, then allows the user to input a partial name and returns the matching restaurant's IPv6 address.

```
Zomato-Style Restaurant Searchrestaurants = {    "Pizza Hut": "2001:db8::1",    "Dominos": "2001:db8::2",    "Burger King": "2001:db8::3",    "McDonalds": "2001:db8::4"}name = input("Enter restaurant name: ").lower()found = Falsefor restaurant, ipv6 in restaurants.items():    if name in restaurant.lower():        print("Restaurant:", restaurant)print("IPv6 Address:", ipv6)        found = Trueif not found:    print ("Restaurant not found!")
```

Example Output

```
Enter restaurant name: pizza

Restaurant: Pizza Hut
IPv6 Address: 2001:db8::1
```

Explanation

1. A dictionary stores restaurant names and their IPv6 addresses.
2. The user enters a full or partial restaurant name.
3. The script searches for matching names without considering uppercase or lowercase.
4. If a match is found, it displays the restaurant name and IPv6 address.
5. If no match is found, it displays `Restaurant not found!`.

5 Use ChatGPT or Copilot to generate a sample IPv6 subnetting scenario (with at least 3 subnets), and then manually assign a valid IPv6 address from one of those subnets to your system. Paste the AI-generated scenario and your configuration steps in a text file.

Practical 5: IPv6 Subnetting and Configuration

1. AI-Generated IPv6 Subnetting Scenario

A college wants to divide its network into three separate subnets for different departments.

Department	IPv6 Subnet	Example Host Address
Administration	`2001:db8:1234:1::/64`	`2001:db8:1234:1::10`
Computer Lab	`2001:db8:1234:2::/64`	`2001:db8:1234:2::10`
Student Wi-Fi	`2001:db8:1234:3::/64`	`2001:db8:1234:3::10`

Note: These are documentation-only IPv6 addresses used to demonstrate subnetting, not for normal internet connectivity.

2. Configuration Steps

Step 1: Open Command Prompt on the Windows laptop.

Step 2: Enter the following command to view the network configuration.

```
ipconfig
```

Step 3: Identify the active network adapter. In my system, the adapter name is `Wi-Fi`.

Step 4: To view the IPv6 addresses assigned to the interfaces, run:

```
netsh interface ipv6 show addresses
```

Step 5: Check the IPv6 configuration using `ipconfig` again after configuring an address from an authorized lab prefix.

3. Actual System Observation

The following details are taken from the `ipconfig` output of my Dell laptop.

Field	Observed Result
Network Adapter	Wi-Fi
IPv6 Address 1	`2401:4900:ae4e:f0ed:731:5380:87bd:435c`
IPv6 Address 2	`2401:4900:ae39:ce7:6b53:1234:7c17:3ec`
Temporary IPv6 Address 1	`2401:4900:ae39:ce7:c0e4:a199:f88d:3f76`
Temporary IPv6 Address 2	`2401:4900:ae4e:f0ed:9863:9389:7e14:f433`
Link-local IPv6 Address	`fe80::ab20:22a6:3b10:1b6%22`
Example Subnet Prefix Length	`/64`
Manual Address Configuration	Not yet completed

4. Conclusion

The AI-generated scenario divides an IPv6 network into three `/64` subnets for Administration, Computer Lab, and Student Wi-Fi. I checked my Dell laptop's Wi-Fi configuration using `ipconfig` and found that IPv6 addresses, including temporary IPv6 addresses, are already assigned. However, a manual address from the example subnet has not yet been configured or verified.